



Society of Petroleum Engineers

SPE-227154-MS

Smart Hydraulic Fracturing: Optimizing Production from Tight Reservoirs

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227154-MS](https://doi.org/10.2118/227154-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

An Operator in Middle East conducts 50+ hydraulic fracturing operations annually in tight reservoirs. Despite the revolutionary impact of hydraulic fracturing, traditional optimization methods are computationally intensive and fail to capture real-world complexities, resulting in production losses. To address this problem, we have developed Smart Hydraulic Fracturing (SHF) optimization, a machine learning solution. SHF forecasts short-term production, and optimizes fracturing and completion parameters to maximize production, addressing operational inefficiencies in hydraulic fracturing.

The solution ingests a large amount of data related to well attributes, well logs, production, fracture design, and completion parameters. The model is trained to forecast short-term production rates or cumulative values for each hydraulic fracturing stage for up to 2-year period. Furthermore, advanced algorithms optimize hydraulic fracture placement and completion parameters towards achieving maximum / minimum objective function for a given well data with multiple stages and reservoir properties. This approach ensures a robust prediction and optimization workflow, streamlining the enhancement of hydraulic fracturing operations without the need for time-consuming, tedious numerical and analytical methods that often involve high uncertainties and lead to unreliable results.

The solution was piloted in one of the operators reservoir involving over 25 wells and 200+ fracturing stages combined. Input data underwent preprocessing, and production was allocated to each stage using data from historical production logging. Subsequently, the model underwent training, testing and validation. The results demonstrated the model's ability to reliably predict production profiles, validated against unseen wells. Furthermore, the fracture design and completion optimization results generated by the solution were reviewed and confirmed by subject matter experts, aligning with historical reservoir and fracturing / completion design experience. The next phase involves scaling the solution across the entire field, training it on a larger number of wells and putting the solution on production to be used by the operator's hydraulic fracturing planning and optimization team.

This paper introduces SHF, a novel machine learning solution tailored for operator's hydraulic fracturing operations in tight reservoirs. The SHF, upon full-scale deployment, will enable the operator to rapidly, accurately, and effectively design and optimize hydraulic fracturing parameters within minutes, a massive

improvement over the conventional method that takes days. Most importantly, it is projected to maximize recovery in challenging tight reservoirs and cutting operational costs through consistently executing optimal hydraulic fracturing operations.

Introduction

The development of tight reservoirs in the Middle East has accelerated over the past decade, driven by national strategies to unlock unconventional resources in regions traditionally dominated by prolific conventional plays. These tight formations—commonly characterized by matrix permeability values below 0.1 mD—require hydraulic fracturing to achieve commercial production. However, compared to North American shale plays, tight reservoirs in the Middle East often present greater geological complexity, including heterogeneous lithofacies, natural fractures, and variable in-situ stress regimes. These conditions reduce the effectiveness of standardized stimulation workflows and necessitate tailored, data-informed approaches.

Despite widespread adoption, conventional hydraulic fracturing design methodologies remain largely empirical or reliant on computationally intensive numerical simulations. These approaches often overlook critical interactions between reservoir heterogeneity, fluid properties, and completion strategies, especially in horizontal wells with multistage completions. The resulting designs can lead to overstimulation, understimulation, or suboptimal stage placement—ultimately impacting production recovery and operational efficiency.

Recent efforts in the Middle East have demonstrated the potential of integrated simulation-based and AI-enhanced workflows to improve stimulation design. [Yang et al. \(2023\)](#) applied a coupled hydraulic fracturing and reservoir simulation model to a carbonate reservoir in the region, showing that optimized designs led to a production increase from approximately 870 to over 2,000 bbl/d. Similarly, [Al-Shuaibi et al. \(2023\)](#) implemented an artificial neural network (ANN)-based optimization workflow in Oman, successfully adapting design parameters to local reservoir and operational constraints.

Globally, machine learning has been used to optimize fracturing in unconventional plays such as shale oil and gas. [Li et al. \(2024\)](#) demonstrated the use of a random forest and particle swarm optimization hybrid model for optimizing fracturing parameters in China's Panke shale oil field, achieving improved forecasting accuracy and economic efficiency. [Ye et al. \(2024\)](#) applied ensemble learning methods to forecast production from hydraulically fractured tight gas wells, reporting high prediction fidelity with an R^2 of up to 0.95. However, these methods often struggle when directly transferred to Middle Eastern contexts due to differences in stress regimes, lithologies, and operational constraints.

In response to these challenges, a collaborative effort has led to the development of the SHF solution. Unlike prior approaches, SHF solution that integrates a machine learning-based production forecasting model with a stage-level optimization engine, enabling rapid, data-driven design of hydraulic fracturing and completion parameters tailored to specific reservoir conditions. The methodology addresses key gaps in the literature by:

- Operating at the fracture stage level rather than the well level;
- Combining forecasting and optimization in a single pipeline;
- Using actual production-logging-derived allocations to enhance model fidelity;
- Delivering results in seconds to minutes, enabling real-time decision support for field operations.

This paper presents the methodology, validation, and deployment results of the SHF solution, and outlines its impact on reducing design time and improving production outcomes across a portfolio of tight reservoir wells.

Methodology

This section outlines the development and deployment of an end-to-end machine learning (ML) solution designed to predict short-term production performance following hydraulic fracturing and to optimize fracturing and completion parameters for multistage wells in tight reservoirs. The methodology consists of two core modules: (1) Production Profile Prediction, and (2) Fracture and Completion Parameter Optimization. These modules work in tandem to deliver a robust and scalable data-driven workflow that minimizes reliance on traditional numerical simulation methods.

Production Profile Prediction Module

The first module focuses on forecasting post-fracturing oil, gas, and water production at the stage level for each well over a 1–2-year horizon. The module leverages supervised machine learning models trained on historical well data to learn the complex, nonlinear relationships between subsurface properties, stimulation design, and production response.

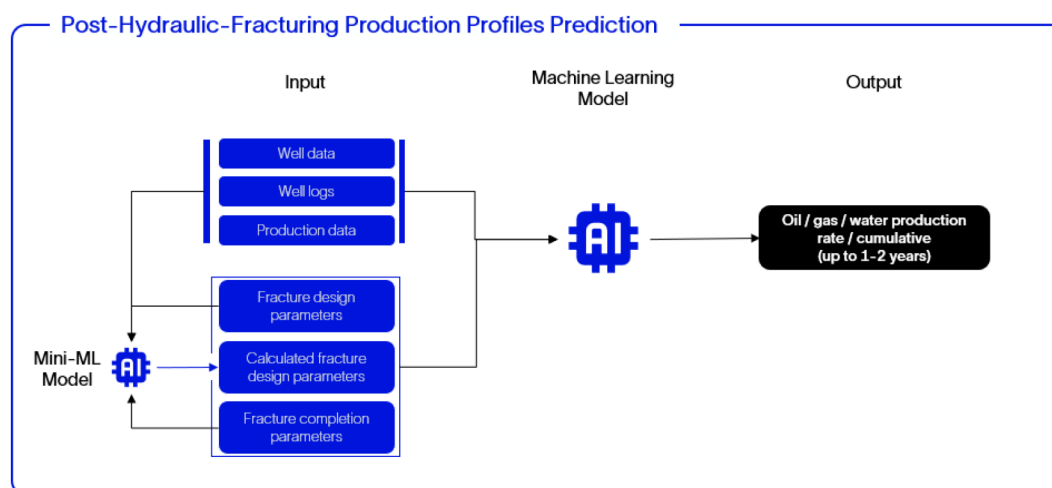


Figure 1—Post-Hydraulic-Fracturing Production Profiles Prediction

Data Inputs. The model is designed to incorporate a wide array of inputs, categorized as follows:

- **Well and Trajectory Data:** Geographic location, azimuth, inclination, and measured depth.
- **Well Logs:** Includes static and dynamic reservoir properties such as porosity, permeability, pore pressure, fluid saturations, net pay thickness, Young's modulus, Poisson's ratio, and minimum horizontal stress.
- **Production Data:** Historical daily or monthly oil, gas, and water rates, cumulative production, production days, tubing head pressure, choke size, and flowing bottomhole pressure.
- **Fracture Design Parameters:** Pumped volumes and types of fluid and proppant, proppant concentration, average fluid rates, and proppant mesh size.
- **Calculated Fracture Properties:** Fracture length, height, and conductivity inferred using a secondary Mini-ML model (see Section 2).
- **Completion Parameters:** Perforation interval and density, number of stages, stage spacing and duration, and hole diameter.

Data Preprocessing Pipeline. A comprehensive preprocessing framework is employed to transform raw engineering and logging data into structured input features for model training. The pipeline includes:

- Automated extraction of well trajectory and location data.

- Computation of statistical descriptors (mean, max, std. dev., etc.) of log data over perforated intervals (i.e., stages).
- Aggregation of fracture design and completion parameters per stage.
- Parsing and integration of historical production log data and allocation of well-level production to individual stages using observed or inferred production logging.

The processed dataset is then used to train the ML model using standard practices of data partitioning, normalization, and feature engineering.

Model Training and Inference. The supervised learning model is trained to predict the production profile for each hydraulic fracturing stage over a 12- to 24-month period. Input features include the preprocessed well, log, completion, and fracture data. The model output comprises time-series predictions of oil, gas, and water production rates or cumulative values for each stage.

During inference, the model is applied to new wells or candidate designs to forecast stage-level production, which is then aggregated to provide overall well-level performance metrics. This enables rapid screening of design alternatives without reliance on physics-based simulation.

Fracture and Completion Parameter Optimization Module

The second module couples the predictive model with a numerical optimization framework to identify the optimal set of hydraulic fracturing and completion parameters that maximize a user-defined objective function.

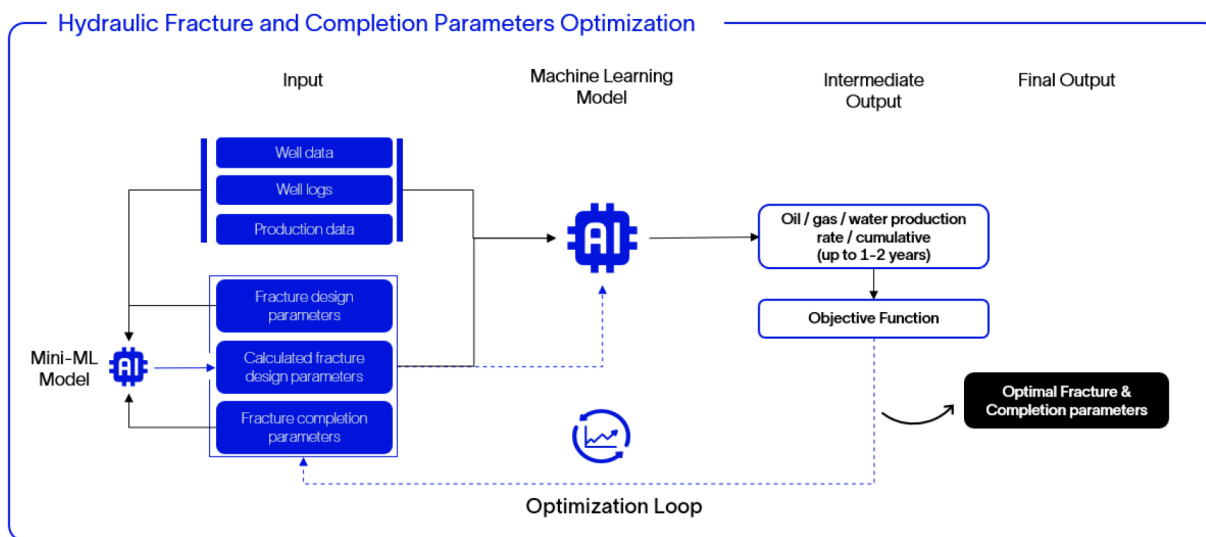


Figure 2—Hydraulic Fracture and Completion Parameters Optimization

Optimization Workflow. The optimization process initiates once the ML production forecasting model is validated. It consists of the following steps:

- **Objective Function Definition:** Users may define objectives such as maximizing cumulative oil production, initial gas rate, or economic metrics like Net Present Value (NPV).
- **Parameter Sampling and Evaluation:** Optimization algorithms (e.g., Differential Evolution, Particle Swarm Optimization) sample across feasible ranges of fracture and completion parameters.
- **Iterative Search:** For each sampled configuration, the ML model predicts the production outcome, which is fed back into the optimizer to refine the parameter search.
- **Optimal Design Output:** The loop continues until convergence is achieved, returning the design that maximizes or minimizes the objective function.

This module transforms the static ML forecast into a dynamic decision-making engine, enabling automated optimization of stimulation designs.

Mini-ML Model for Fracture Geometry Prediction. To ensure complete design inputs without reliance on empirical or simulation-based fracture geometry models, a supplementary Mini-ML model is embedded into the workflow. This model predicts fracture length, height, and conductivity based on historical data, well logs, and design parameters.

These properties—typically inferred via diagnostic fracture injection tests (DFITs) or simulated via numerical models—are instead generated in real time using machine learning, enhancing both the prediction accuracy and optimization robustness.

Case Study

To evaluate the performance and scalability of the SHF solution, a multi-step case study was conducted using historical data from a tight reservoir field in the Middle East. The objective was to assess the model's ability to forecast short-term post-fracture production and optimize completion designs across multistage horizontal wells.

Model Development and Iterative Training

The machine learning model was developed in an iterative manner, with a progressively increasing volume of training data in each phase. This allowed for refinement of data preprocessing workflows, model selection, and performance evaluation.

Phase 1: Initial Feasibility (3 Wells, ~20 Stages). The first iteration involved data from 3 wells, covering over 20 hydraulic fracturing stages. This phase served as a feasibility study to validate the modeling approach and the effectiveness of the feature engineering strategy. Despite the limited data, early results confirmed the potential for capturing production trends at the stage level.

Phase 2: Intermediate Training (10 Wells, ~80 Stages). In the second phase, data from 10 wells—representing more than 80 fracturing stages—were used to train and validate the model. A range of ensemble models were tested, and the Extremely Randomized Trees Regressor (ExtraTrees) demonstrated the best performance. The model achieved an R^2 score of 0.65 when tested on blind well X, indicating a strong predictive capability given the available dataset.

Phase 3: Full Pilot Model (25 Wells, ~150 Stages). The final phase of the pilot involved training the model on 25 wells, comprising over 150 individual stages. The same ExtraTrees model was retained for consistency, and it achieved an R^2 score of 0.85 on the validation dataset. Better than in Phase 2, this increase is attributed to increased data diversity and inclusion of more heterogeneous well conditions. Importantly, the model continued to show improved generalizability, evidenced by increased performance when tested on blind well Y (R^2 improved from 0.65 to 0.85 as the training set grew). This highlights the importance of data volume in enhancing model accuracy and robustness.

Machine Learning Model Algorithm - Example. The machine learning model used for production prediction in this study is based on the Extremely Randomized Trees (ExtraTrees) algorithm, a variant of ensemble learning methods that enhances model robustness by introducing additional layers of randomness. The full prediction workflow is illustrated in [Fig 3](#).

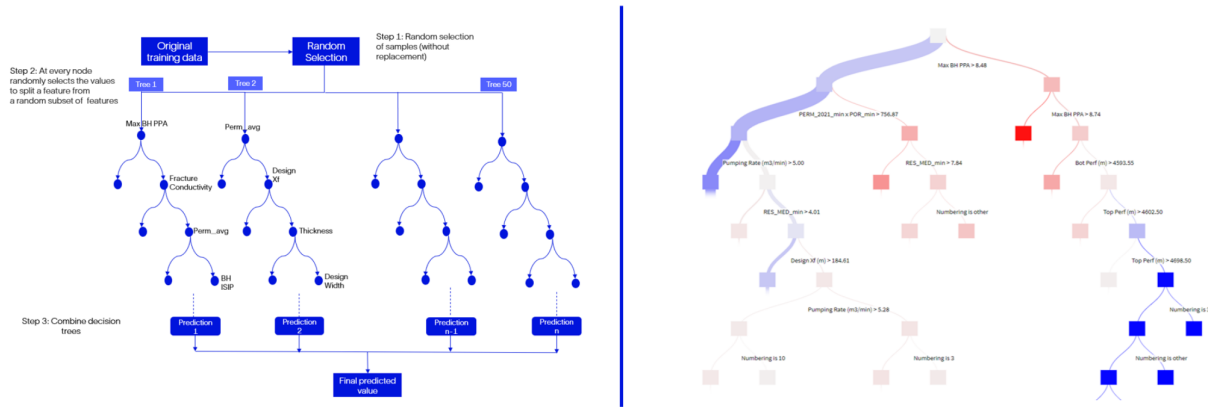


Figure 3—High level illustration of decision making of ExtraTrees ML algorithm

As shown in the figure, the model constructs an ensemble of decision trees, each trained on a randomly selected subset of the original training data (Step 1). Within each tree, node-level splits are performed using a randomly chosen subset of input features (Step 2), allowing for diverse decision paths across trees. For example, Tree 1 utilizes features such as *Max Bottomhole PPA*, *Fracture Conductivity*, *Average Permeability*, and *Bottomhole ISIP*, while Tree 2 is built on different features like *Perm_avg*, *Design Fracture Half-Length (Xf)*, *Reservoir Thickness*, and *Design Width*. Other trees in the ensemble follow similar logic, each exploring different combinations of available input features.

After the trees independently produce their respective predictions—each reflecting a unique view of the data—these outputs are aggregated (Step 3), typically through averaging, to generate the final predicted value for short-term oil, gas, or water production. This ensemble approach captures complex, nonlinear interactions among geological, geomechanical, and design parameters, and enhances model accuracy while mitigating overfitting, even in datasets with limited well and stage-level data.

Results

Stage-Level Design Optimization and Production Improvement. The SHF solution was deployed on a two-stage well to evaluate its ability to optimize hydraulic fracturing and completion parameters for improved production outcomes. The comparative input data for the base and optimized scenarios are shown in Fig 5, followed by the associated production forecasts in Fig 4.

Base Scenario Data

Top Perforation (m)	Bottom Perforation (m)	Bottomhole Closure Pressure (psi)	Pumping Rate (m3/min)	Design Width (mm)	Design XF (m)	Design H (m)	Fracture Conductivity (mD.ft)	Bottomhole ISIP (psi)	PAD Actual (M3)	Total Volume (m3)	Maximum Bottomhole PPA	Propan in Formation (KG)	Actions
XX34.5	XX71.5	10095.49	4.5	7.3411	128.0822	37.3809	2528.9247	11530.24	96.59	287.62	5.7089	61600.88	 
XX08.3	XX45.8	10095.49	4.3536	7.3411	128.0822	37.3809	2528.9247	11530.24	96.59	287.62	5.7089	61600.88	 

Optimized Scenario Data

Stage	Top Perforation (m)	Bottom Perforation (m)	Pumping Rate (m3/min)	Design Width (mm)	Design XF (m)	Design H (m)	Fracture Conductivity (mD.ft)	PAD Actual (M3)	Total Volume (m3)	Maximum Bottomhole PPA	Propan in Formation (KG)	Optimal Annual Production
1	XX34.5	XX71.5	5.5	5	186	59	8282	221	958	7	119928	21537157.6737
2	XX08.3	XX45.8	6	11	212	62	6446	80	925	6	171895	16220265.487

Figure 4—Input data for Base Scenario and Optimized parameters for Optimized Scenario

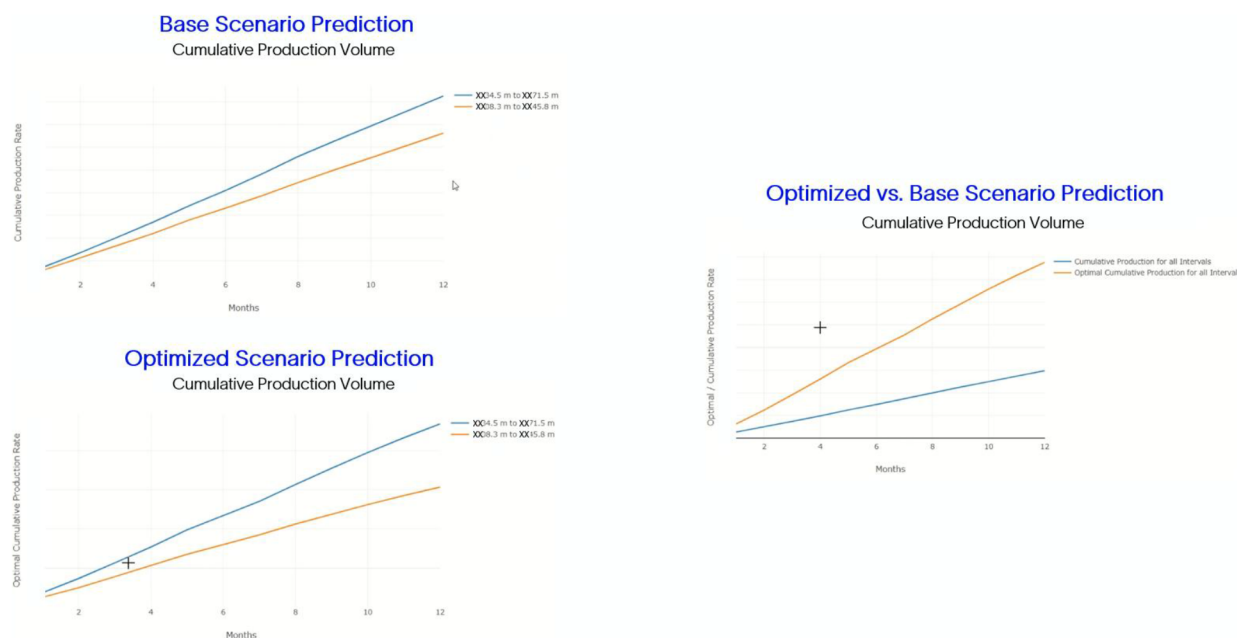


Figure 5—Base and Optimized Scenario Cumulative Production Comparison

Design Parameter Comparison. In the base case, both stages utilized a uniform design with modest pumping rates (~ 4.5 m³/min), consistent design half-lengths ($X_f = 128$ m), and relatively low fracture conductivity ($\sim 2,529$ mD·ft). The total fluid volume was ~ 287 m³ per stage, and proppant placement remained constant at $\sim 61,600$ kg.

In contrast, the optimized scenario exhibited significant design improvements across both stages. Pumping rates were increased to 5.5 and 6.0 m³/min, and design parameters were dynamically adjusted. For instance, Stage 1 was configured with $X_f = 186$ m, $H = 59$ m, and fracture conductivity of 8,282 mD·ft—over $3\times$ higher conductivity than the base case. The optimized design also featured varied proppant loading and total volumes tailored to each interval's unique characteristics, resulting in significantly enhanced stimulation geometry and conductivity.

Production Forecast Comparison. The resulting production profiles are presented in Fig 5, which includes:

- Top left: Base scenario cumulative production volume by stage
- Bottom left: Optimized scenario cumulative production volume by stage
- Right: Aggregated cumulative production across both stages—base vs. optimized

In the base case, Stage 1 (XX34.5 m to XX71.5 m) outperformed Stage 2 marginally. However, both stages exhibited similar gradients.

Post-optimization, both stages showed marked improvements, representing a $3\times$ to $4\times$ increase in output. The optimized design leveraged higher injection rates and tailored design geometries, yielding more efficient fracture networks and greater flow capacity.

The cumulative comparison in the right panel of Fig 5 clearly demonstrates the uplift: the optimized scenario significantly outpaces the base case in total production across all months, with the gap widening over time. This highlights the effectiveness of the SHF system's optimization engine in identifying superior design configurations that drive tangible production increases.

These gains were achieved through the optimization of stage-level fracture and completion parameters such as fluid volumes, proppant concentration, stage spacing, and perforation density. The ML model

evaluated numerous design permutations using the trained machine learning model and selected the configuration that maximized cumulative oil output over the defined horizon.

Key Observations and Insights

- **Impact of Training Data Volume:** Model accuracy consistently improved with the inclusion of more wells and stages. Increasing the training dataset from 10 to 25 wells led to a noticeable improvement in blind-well predictive accuracy, underscoring the sensitivity of machine learning performance to dataset size.
- **Data Allocation Strategy:** Accurate production allocation at the stage level using production logs (e.g., MPLT) significantly contributed to model precision.
- **Data Gaps and Quality:** Certain datasets contained incomplete information (e.g., missing QGZI rates), which limited their utility during model training. Filling these gaps is essential to improving the training coverage and maximizing the potential of the available data.

Way Forward

Looking ahead, several strategic enhancements are planned to improve the scalability, robustness, and operational impact of the SHF solution.

A key priority is the expansion of the training dataset. While current results validate the model's potential, large-scale training on hundreds of wells and thousands of stages is essential to improve generalization and support cross-field adaptability. With a more diverse and representative dataset, the model is expected to better capture variations in lithology, reservoir quality, and stimulation practices, leading to more reliable predictions across different assets.

Model refinement is another critical area. Future development will focus on improving prediction accuracy in edge cases—particularly in depleted pressure zones or stages with unusual reservoir behavior. Computational efficiency will also be enhanced to enable real-time or near-real-time predictions and optimization, which is crucial for integration into daily field operations. In addition, augmenting the feature set with new inputs such as stimulation pressure data, geomechanical parameters, or time-lapse diagnostic results is anticipated to improve the model's explanatory power.

On the optimization side, ongoing work aims to broaden the scope of the objective function beyond production volume alone. Economic indicators such as Net Present Value (NPV) and cost-effectiveness will be incorporated to reflect business objectives more holistically. Furthermore, the optimization framework will be extended to support multi-objective scenarios, enabling balanced trade-offs between competing goals like production maximization, cost reduction, and operational constraints. Advanced optimization algorithms, particularly those suited for high-dimensional and nonlinear design spaces, will be evaluated and integrated as needed.

Beyond data and algorithmic improvements, efforts are underway to standardize and streamline the data ingestion pipeline. Current workflows require customization due to inconsistencies in data formats—especially from sources such as MPLT logs and completion reports. Establishing a unified parsing and preprocessing framework will reduce manual intervention and support a more generalized, scalable deployment.

Lastly, user accessibility and operationalization of the solution remain critical. Plans are in place to embed SHF within a web-based interface or existing engineering platforms, allowing domain experts and field engineers to interact seamlessly with the system. This transition from pilot to field-scale deployment will mark a major milestone in delivering on the promise of smart, data-driven hydraulic fracturing design.

Conclusions

This study presents the development and application of the SHF solution—an integrated machine learning-based framework designed to forecast post-fracture production and optimize hydraulic fracturing and completion parameters in tight reservoirs. Unlike conventional approaches, SHF delivers rapid, stage-level predictions and data-driven design recommendations without requiring time-consuming numerical simulations or oversimplified analytical models.

The solution was piloted using a progressively scaled training dataset comprising 25 wells and over 150 hydraulic fracturing stages. The selected ExtraTrees regression model demonstrated consistent improvement in predictive performance with increased training data, achieving an R^2 score of up to 0.85 on blind-well evaluation. By integrating production forecasting and optimization in a unified pipeline, SHF enables engineers to make faster, more informed decisions that align with field-specific production behaviors.

The successful pilot showcases the technical feasibility and operational value of deploying machine learning to support hydraulic fracturing design in complex tight reservoirs, particularly within the Middle East context where conventional design workflows remain resource-intensive and uncertain.

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